Activity_Validate and clean your data

April 13, 2026

1 Activity: Validate and clean your data

1.1 Introduction

In this activity, you will use input validation and label encoding to prepare a dataset for analysis. These are fundamental techniques used in all types of data analysis, from simple linear regression to complex neural networks.

In this activity, you are a data professional an investment firm that is attempting to invest in private companies with a valuation of at least $1 billion. These are often known as "unicorns." Your client wants to develop a better understanding of unicorns, with the hope they can be early investors in future highly successful companies. They are particularly interested in the investment strategies of the three top unicorn investors: Sequoia Capital, Tiger Global Management, and Accel.

1.2 Step 1: Imports

Import relevant Python libraries and packages: `numpy`, `pandas`, `seaborn`, and `pyplot` from `matplotlib`.

[118]:
```
# Import libraries and packages.

### YOUR CODE HERE ###
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import datetime as dt
```

1.2.1 Load the dataset

The data contains details about unicorn companies, such as when they were founded, when they achieved unicorn status, and their current valuation. The dataset `Modified_Unicorn_Companies.csv` is loaded as `companies`, now display the first five rows. The variables in the dataset have been adjusted to suit the objectives of this lab, so they may be different from similar data used in prior labs. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

[119]:
```python
# Run this cell so pandas displays all columns
pd.set_option('display.max_columns', None)
```

[120]:
```python
# RUN THIS CELL TO IMPORT YOUR DATA.
companies = pd.read_csv('Modified_Unicorn_Companies.csv')

# Display the first five rows.
### YOUR CODE HERE ###
companies.head(5)
```

[120]:
```
     Company  Valuation Date Joined                         Industry  \
0  Bytedance        180  2017-04-07          Artificial intelligence
1     SpaceX        100  2012-12-01                            Other
2      SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
3     Stripe         95  2014-01-23                          FinTech
4     Klarna         46  2011-12-12                          Fintech

            City Country/Region      Continent  Year Founded Funding  \
0        Beijing          China           Asia          2012     $8B
1      Hawthorne  United States  North America          2002     $7B
2       Shenzhen          China           Asia          2008     $2B
3  San Francisco  United States  North America          2010     $2B
4      Stockholm         Sweden         Europe          2005     $4B

                                    Select Investors
0  Sequoia Capital China, SIG Asia Investments, S…
1  Founders Fund, Draper Fisher Jurvetson, Rothen…
2  Tiger Global Management, Sequoia Capital China…
3        Khosla Ventures, LowercaseCapital, capitalG
4  Institutional Venture Partners, Sequoia Capita…
```

1.3 Step 2: Data cleaning

Begin by displaying the data types of the columns in `companies`.

[121]:
```python
# Display the data types of the columns.

### YOUR CODE HERE ###
companies.dtypes
```

[121]:
```
Company             object
Valuation            int64
Date Joined         object
Industry            object
```

```
City                  object
Country/Region        object
Continent             object
Year Founded           int64
Funding               object
Select Investors      object
dtype: object
```

Hint 1

Review what you have learned about exploratory data analysis in Python.

Hint 2

There is a `pandas` DataFrame property that displays the data types of the columns in the specified DataFrame.

Hint 3

The `pandas` DataFrame `dtypes` property will be helpful.

1.3.1 Modify the data types

Notice that the data type of the `Date Joined` column is an `object`—in this case, a string. Convert this column to `datetime` to make it more usable.

[122]:
```python
# Apply necessary datatype conversions.

### YOUR CODE HERE ###
companies.head(1)
companies['Date Joined']=pd.to_datetime(companies['Date Joined'])
companies.dtypes
```

[122]:
```
Company                       object
Valuation                      int64
Date Joined           datetime64[ns]
Industry                      object
City                          object
Country/Region                object
Continent                     object
Year Founded                   int64
Funding                       object
Select Investors              object
dtype: object
```

[123]:
```python
companies.head()
```

[123]:
```
     Company  Valuation Date Joined                         Industry  \
0  Bytedance        180  2017-04-07          Artificial intelligence
```

```
1     SpaceX        100  2012-12-01                             Other
2      SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
3     Stripe         95  2014-01-23                           FinTech
4     Klarna         46  2011-12-12                           Fintech

            City Country/Region      Continent  Year Founded Funding  \
0        Beijing          China           Asia          2012     $8B
1      Hawthorne  United States  North America          2002     $7B
2       Shenzhen          China           Asia          2008     $2B
3  San Francisco  United States  North America          2010     $2B
4      Stockholm         Sweden         Europe          2005     $4B

                                    Select Investors
0  Sequoia Capital China, SIG Asia Investments, S…
1  Founders Fund, Draper Fisher Jurvetson, Rothen…
2  Tiger Global Management, Sequoia Capital China…
3        Khosla Ventures, LowercaseCapital, capitalG
4  Institutional Venture Partners, Sequoia Capita…
```

[124]:
```
companies['Year Joined']=companies['Date Joined'].dt.year
```

[125]:
```
companies.head()
```

[125]:
```
     Company  Valuation Date Joined                         Industry  \
0  Bytedance        180  2017-04-07          Artificial intelligence
1     SpaceX        100  2012-12-01                            Other
2      SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
3     Stripe         95  2014-01-23                          FinTech
4     Klarna         46  2011-12-12                          Fintech

            City Country/Region      Continent  Year Founded Funding  \
0        Beijing          China           Asia          2012     $8B
1      Hawthorne  United States  North America          2002     $7B
2       Shenzhen          China           Asia          2008     $2B
3  San Francisco  United States  North America          2010     $2B
4      Stockholm         Sweden         Europe          2005     $4B

                                    Select Investors  Year Joined
0  Sequoia Capital China, SIG Asia Investments, S…        2017
1  Founders Fund, Draper Fisher Jurvetson, Rothen…        2012
2  Tiger Global Management, Sequoia Capital China…        2018
3        Khosla Ventures, LowercaseCapital, capitalG        2014
4  Institutional Venture Partners, Sequoia Capita…        2011
```

1.3.2 Create a new column

Add a column called `Years To Unicorn`, which is the number of years between when the company was founded and when it became a unicorn.

[126]:
```
# Create the column Years To Unicorn.

### YOUR CODE HERE ###
companies['Years To unicorn']=companies['Year Joined']-companies['Year Founded']
```

[127]:
```
companies.head(2)
```

[127]:
```
     Company  Valuation Date Joined                Industry       City  \
0  Bytedance        180  2017-04-07  Artificial intelligence    Beijing
1     SpaceX        100  2012-12-01                    Other  Hawthorne

  Country/Region      Continent  Year Founded Funding  \
0          China           Asia          2012     $8B
1  United States  North America          2002     $7B

                                     Select Investors  Year Joined  \
0  Sequoia Capital China, SIG Asia Investments, S…         2017
1  Founders Fund, Draper Fisher Jurvetson, Rothen…         2012

   Years To unicorn
0                 5
1                10
```

Hint 1

Extract just the year from the `Date Joined` column.

Hint 2

Use `dt.year` to access the year of a datetime object.

Hint 3

Subtract the `Year Founded` from the `Date Joined`, and save it to a new column called `Years To Unicorn`.

Ensure you're properly extracting just the year (as an integer) from `Date Joined`.

QUESTION: Why might your client be interested in how quickly a company achieved unicorn status?

[Write your response here. Double-click (or enter) to edit.]

1.3.3 Input validation

The data has some issues with bad data, duplicate rows, and inconsistent `Industry` labels.

Identify and correct each of these issues.

Correcting bad data Get descriptive statistics for the `Years To Unicorn` column.

[128]:
```
# Identify and correct the issue with Years To Unicorn.

### YOUR CODE HERE ###
companies.describe()
```

[128]:
```
         Valuation  Year Founded  Year Joined  Years To unicorn
count  1074.000000   1074.000000  1074.000000       1074.000000
mean      3.445996   2012.870577  2019.883613          7.013035
std       8.544242      5.705494     2.008011          5.331842
min       1.000000   1919.000000  2007.000000         -3.000000
25%       1.000000   2011.000000  2019.000000          4.000000
50%       2.000000   2014.000000  2021.000000          6.000000
75%       3.000000   2016.000000  2021.000000          9.000000
max     180.000000   2021.000000  2022.000000         98.000000
```

Hint 1

Use the `describe()` method on the series. Considering the results, does anything seem problematic?

Hint 2

A company cannot reach unicorn status before it is founded. In other words, `Years to Unicorn` cannot be less than 0.

Hint 3

Using the `describe()` method on the `Years To Unicorn` series shows that the minimum value is `-3`. Since there cannot be negative time, this value and possibly others are problematic.

Isolate all rows where the `Years To Unicorn` column contains a negative value.

[129]:
```
# Isolate any rows where `Years To Unicorn` is negative

### YOUR CODE HERE ###
filtered = companies[companies['Years To unicorn']<0]
filtered
```

[129]:
```
      Company  Valuation Date Joined                      Industry      City  \
527  InVision          2  2017-11-01  Internet software & services  New York

    Country/Region      Continent  Year Founded Funding  \
527  United States  North America          2020    $349M

                                        Select Investors  Year Joined  \
527  FirstMark Capital, Tiger Global Management, IC…         2017
```

```
     Years To unicorn
527                -3
```

Question: How many rows have negative values in the `Years To Unicorn` column, and what companies are they for?

There is one row which has a negative value in it.

An internet search reveals that InVision was founded in 2011. Replace the value at `Year Founded` with `2011` for InVision's row.

[130]:
```
# Replace InVision's `Year Founded` value with 2011
companies['Year Founded']=companies['Year Founded'].replace("2020","2011")
### YOUR CODE HERE ###

# Verify the change was made proper

### YOUR CODE HERE ###
companies.iloc[527]
```

[130]:
```
Company                                                  InVision
Valuation                                                       2
Date Joined                                   2017-11-01 00:00:00
Industry                                Internet software & services
City                                                     New York
Country/Region                                      United States
Continent                                           North America
Year Founded                                                 2020
Funding                                                     $349M
Select Investors    FirstMark Capital, Tiger Global Management, IC…
Year Joined                                                  2017
Years To unicorn                                               -3
Name: 527, dtype: object
```

Hint 1

To overwrite data in a dataframe in a situation like this, you should use `loc[]` or `iloc[]` selection statements. Otherwise, you might overwrite to a view of the dataframe, which means that you're not overwriting the data in the dataframe itself, and the change will not take permanent effect.

Hint 2

The following code will **not** work:

companies[companies['Company']=='InVision']['Year Founded'] = 2011

You must use either `loc[]` or `iloc[]`.

Now, recalculate all the values in the `Years To Unicorn` column to remove the negative value for InVision. Verify that there are no more negative values afterwards.

[131]:
```
# Recalculate all values in the `Years To Unicorn` column

### YOUR CODE HERE ###
companies.head(10)
# Verify that there are no more negative values in the column
filtered=companies[companies['Years To unicorn']<0]
### YOUR CODE HERE ###
filtered.isnull()
```

[131]:
```
     Company  Valuation  Date Joined  Industry   City  Country/Region  \
527    False      False        False     False  False           False

     Continent  Year Founded  Funding  Select Investors  Year Joined  \
527      False         False    False             False        False

     Years To unicorn
527             False
```

Issues with Industry labels The company provided you with the following list of industry labels to identify in the data for `Industry`.

Note: Any labels in the `Industry` column that are not in `industry_list` are misspellings.

[132]:
```
# List provided by the company of the expected industry labels in the data
industry_list = ['Artificial intelligence', 'Other','E-commerce &␣
↪direct-to-consumer', 'Fintech',\
      'Internet software & services','Supply chain, logistics, & delivery',␣
↪'Consumer & retail',\
      'Data management & analytics', 'Edtech', 'Health', 'Hardware','Auto &␣
↪transportation', \
       'Travel', 'Cybersecurity','Mobile & telecommunications']
```

[133]:
```
industry_set=set(industry_list)
```

First, check if there are values in the `Industry` column that are not in `industry_list`. If so, what are they?

[134]:
```
# Check which values are in `Industry` but not in `industry_list`

### YOUR CODE HERE ###
companies.head(2)
companies.dtypes
```

[134]:
```
Company                     object
Valuation                    int64
Date Joined         datetime64[ns]
Industry                    object
```

```
City                      object
Country/Region            object
Continent                 object
Year Founded               int64
Funding                   object
Select Investors          object
Year Joined                int64
Years To unicorn           int64
dtype: object
```

[135]:
```python
my_set=set(companies['Industry'])
```

[136]:
```python
missing_set=my_set-industry_set
```

[137]:
```python
missing_set
```

[137]:
```
{'Artificial Intelligence', 'Data management and analytics', 'FinTech'}
```

HINT 1

There are many ways to do this. One approach is to consider what data type reduces iterable sequences to their unique elements and allows you to compare membership.

HINT 2

A set is a data type that consists of unique elements and supports membership testing with other sets.

HINT 3

Set A – Set B will result in all the elements that are in Set A but not in Set B. Convert `industry_list` to a set and subtract it from the set of the values in the `Industry` series.

Question: Which values currently exist in the `Industry` column that are not in `industry_list`?

{'Artificial Intelligence', 'Data management and analytics', 'FinTech'} these are the values that are missing/incorrect values or data .Note that data is similar but however the capitalization has made a difference over here .

[ ]:
```python

```

Now, correct the bad entries in the `Industry` column by replacing them with an approved string from `industry_list`. To do this, use the `replace()` `Series` method on the `Industry` series. When you pass a dictionary to the method, it will replace the data in the series where that data matches the dictionary's keys. The values that get imputed are the values of the dictionary. If a value is not specified in the dictionary, the series' original value is retained.

`Example:`

```
[IN]: column_a = pd.Series(['A', 'B', 'C', 'D'])
      column_a
```

```
[OUT]: 0    A
       1    B
       2    C
       3    D
       dtype: object

 [IN]: replacement_dict = {'A':'z', 'B':'y', 'C':'x'}
       column_a = column_a.replace(replacement_dict)
       column_a

[OUT]: 0    z
       1    y
       2    x
       3    D
       dtype: object
```

1. Create a dictionary called `replacement_dict` whose keys are the incorrect spellings in the `Industry` series and whose values are the correct spelling, as indicated in `industry_list`.
2. Call the `replace()` method on the `Industry` series and pass to it `replacement_dict` as its argument. Reassign the result back to the `Industry` column.
3. Verify that there are no longer any elements in `Industry` that are not in `industry_list`.

```
[138]: # 1. Create `replacement_dict`

       ### YOUR CODE HERE ###
       replacement_dict={'Artificial Intelligence':'Artificial intelligence','Data␣
        ↪management and analytics':'Data management & analytics','FinTech':'Fintech'}
       # 2. Replace the incorrect values in the `Industry` column
       companies['Industry']=companies['Industry'].replace(replacement_dict)
       ### YOUR CODE HERE ###
       companies.head(3)
       # 3. Verify that there are no longer any elements in `Industry` that are not in␣
        ↪`industry_list`

       ### YOUR CODE HERE ###
```

```
[138]:     Company  Valuation Date Joined                       Industry  \
       0  Bytedance        180  2017-04-07         Artificial intelligence
       1     SpaceX        100  2012-12-01                           Other
       2      SHEIN        100  2018-07-03  E-commerce & direct-to-consumer

               City Country/Region      Continent  Year Founded Funding  \
       0    Beijing          China           Asia          2012     $8B
       1  Hawthorne  United States  North America          2002     $7B
       2   Shenzhen          China           Asia          2008     $2B
```

```
                                 Select Investors  Year Joined  \
0  Sequoia Capital China, SIG Asia Investments, S…         2017
1  Founders Fund, Draper Fisher Jurvetson, Rothen…         2012
2  Tiger Global Management, Sequoia Capital China…         2018

   Years To unicorn
0                 5
1                10
2                10
```

[139]:
```
checkset=set(companies['Industry'])
check_set=checkset-industry_set
check_set
```

[139]:
```
set()
```

[140]:
```
# We have successfully fixed the issue of incorrect/inconsistent values or data␣
↪in our dataset
```

Hint 1

Refer to the example provided for how to use the `replace()` `Series` method.

Hint 2

When you define the `replacement_dict` dictionary, the misspellings should be the keys and the correct spellings should be the values.

Hint 3

When you call `replace()` on the `Industry` series and pass to it the `replacement_dict` dictionary as an argument, you must reassign the result back to `companies['Industry']` for the change to take effect.

Handling duplicate rows The business mentioned that no company should appear in the data more than once.

Verify that this is indeed the case, and if not, clean the data so each company appears only once.

Begin by checking which, if any, companies are duplicated. Filter the data to return all occurrences of those duplicated companies.

[141]:
```
# Isolate rows of all companies that have duplicates

### YOUR CODE HERE ###
companies['data']=companies['Company'].duplicated(keep=False)

filter1=companies[companies['data']==True]
filter1
```

```
[141]:         Company  Valuation Date Joined                  Industry         City  \
385       BrewDog          2  2017-04-10         Consumer & retail     Aberdeen
386       BrewDog          2  2017-04-10         Consumer & retail     Aberdeen
510        ZocDoc          2  2015-08-20                    Health     New York
511        ZocDoc          2  2015-08-20                    Health          NaN
1031   SoundHound          1  2018-05-03   Artificial intelligence  Santa Clara
1032   SoundHound          1  2018-05-03                     Other  Santa Clara

       Country/Region      Continent  Year Founded Funding  \
385    United Kingdom         Europe          2007   $233M
386     UnitedKingdom         Europe          2007   $233M
510     United States  North America          2007   $374M
511     United States  North America          2007   $374M
1031    United States  North America          2005   $215M
1032    United States  North America          2005   $215M

                                        Select Investors  Year Joined  \
385                     TSG Consumer Partners, Crowdcube         2017
386                                TSG Consumer Partners         2017
510       Founders Fund, Khosla Ventures, Goldman Sachs         2015
511                                        Founders Fund         2015
1031   Tencent Holdings, Walden Venture Capital, Glob…        2018
1032                                    Tencent Holdings         2018

       Years To unicorn  data
385                  10  True
386                  10  True
510                   8  True
511                   8  True
1031                 13  True
1032                 13  True
```

```
[ ]:
```

Check for duplicated values specifically in the `Company` column, not entire rows that are duplicated.

Hint 2

The pandas `duplicated()` `DataFrame` method can indentify duplicated rows. Apply it to the `Company` column in `companies` to find which companies appear more than once.

Hint 3

- To specify that you want to check for duplicates only in the `Company` column, indicate this with the `subset` parameter.
- To return *all* occurrences of duplicates, set the `keep` parameter to `False`.

Question: Do these duplicated companies seem like legitimate data points, or are they problematic data?

[Write your response here. Double-click (or enter) to edit.]

Keep the first occurrence of each duplicate company and drop the subsequent rows that are copies.

[142]:
```
# Drop rows of duplicate companies after their first occurrence

### YOUR CODE HERE ###
companies=companies.drop_duplicates(subset='Company')
companies['data']=companies['Company'].duplicated(keep=False)
filter3=companies[companies['data']==True]
filter3
filter3.sum()
```

[142]:
```
Company             0.0
Valuation           0.0
Industry            0.0
City                0.0
Country/Region      0.0
Continent           0.0
Year Founded        0.0
Funding             0.0
Select Investors    0.0
Year Joined         0.0
Years To unicorn    0.0
data                0.0
dtype: float64
```

[143]:
```
# Therefore over here we have successfully removed all the duplicate values.
```

Hint 1

Use the `drop_duplicates()` `DataFrame` method.

Hint 2

Make sure to subset `Company` and reassign the results back to the `companies` dataframe for the changes to take effect.

Question: Why is it important to perform input validation?

Inorder to ensure that dataset is consistent ,clean and effective

Question: What steps did you take to perform input validation for this dataset?

removed all the duplicate values and also performed many other steps with it.

1.3.4 Convert numerical data to categorical data

Sometimes, you'll want to simplify a numeric column by converting it to a categorical column. To do this, one common approach is to break the range of possible values into a defined number of equally sized bins and assign each bin a name. In the next step, you'll practice this process.

Create a `High Valuation` column The data in the `Valuation` column represents how much money (in billions, USD) each company is valued at. Use the `Valuation` column to create a new column called `High Valuation`. For each company, the value in this column should be `low` if the company is in the bottom 50% of company valuations and `high` if the company is in the top 50%.

[144]:
```
# Create new `High Valuation` column
# Use qcut to divide Valuation into 'high' and 'low' Valuation groups
companies.head(2)

companies['High Valuation']=pd.
↪qcut(companies['Valuation'],2,labels=['low','high'])
### YOUR CODE HERE ###
companies.iloc[300]
```

[144]:
```
Company                                                          Graphcore
Valuation                                                                3
Date Joined                                            2018-12-18 00:00:00
Industry                                           Artificial intelligence
City                                                               Bristol
Country/Region                                              United Kingdom
Continent                                                           Europe
Year Founded                                                          2016
Funding                                                              $682M
Select Investors    Dell Technologies Capital, Pitango Venture Cap…
Year Joined                                                           2018
Years To unicorn                                                         2
data                                                                 False
High Valuation                                                        high
Name: 300, dtype: object
```

Hint 1

There are multiple ways to complete this task. Review what you've learned about organizing data into equal quantiles.

Hint 2

Consider using the pandas `qcut()` function.

Hint 3

Use `pandas qcut()` to divide the data into two equal-sized quantile buckets. Use the `labels` parameter to define the output labels. The values you give for `labels` will be the values that are inserted into the new column.

1.3.5 Convert categorical data to numerical data

Three common methods for changing categorical data to numerical are:

1. Label encoding: order matters (ordinal numeric labels)

2. Label encoding: order doesn't matter (nominal numeric labels)
3. Dummy encoding: order doesn't matter (creation of binary columns for each possible category contained in the variable)

The decision on which method to use depends on the context and must be made on a case-to-case basis. However, a distinction is typically made between categorical variables with equal weight given to all possible categories vs. variables with a hierarchical structure of importance to their possible categories.

For example, a variable called `subject` might have possible values of `history`, `mathematics`, `literature`. In this case, each subject might be **nominal**—given the same level of importance. However, you might have another variable called `class`, whose possible values are `freshman`, `sophomore`, `junior`, `senior`. In this case, the class variable is **ordinal**—its values have an ordered, hierarchical structure of importance.

Machine learning models typically need all data to be numeric, and they generally use ordinal label encoding (method 1) and dummy encoding (method 3).

In the next steps, you'll convert the following variables: `Continent`, `Country/Region`, and `Industry`, each using a different approach.

1.3.6 Convert `Continent` to numeric

For the purposes of this exercise, suppose that the investment group has specified that they want to give more weight to continents with fewer unicorn companies because they believe this could indicate unrealized market potential.

Question: Which type of variable would this make the `Continent` variable in terms of how it would be converted to a numeric data type?

[Write your response here. Double-click (or enter) to edit.]

[145]:
```python
companies.head(4)
```

[145]:
```
     Company  Valuation Date Joined                          Industry  \
0  Bytedance        180  2017-04-07           Artificial intelligence
1     SpaceX        100  2012-12-01                             Other
2      SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
3     Stripe         95  2014-01-23                           Fintech

            City Country/Region      Continent  Year Founded Funding  \
0        Beijing          China           Asia          2012     $8B
1      Hawthorne  United States  North America          2002     $7B
2       Shenzhen          China           Asia          2008     $2B
3  San Francisco  United States  North America          2010     $2B

                                   Select Investors  Year Joined  \
0  Sequoia Capital China, SIG Asia Investments, S…         2017
1  Founders Fund, Draper Fisher Jurvetson, Rothen…         2012
2  Tiger Global Management, Sequoia Capital China…         2018
```

```
3       Khosla Ventures, LowercaseCapital, capitalG        2014

   Years To unicorn   data High Valuation
0                 5  False           high
1                10  False           high
2                10  False           high
3                 4  False           high
```

Rank the continents in descending order from the greatest number of unicorn companies to the least.

[146]:
```
# Rank the continents by number of unicorn companies

### YOUR CODE HERE #
rank=companies['Continent'].value_counts()
rank
```

[146]:
```
North America    586
Asia             310
Europe           143
South America     21
Oceania            8
Africa             3
Name: Continent, dtype: int64
```

Hint

Use the `value_counts()` method on the `Continent` series.

Now, create a new column called `Continent Number` that represents the `Continent` column converted to numeric in the order of their number of unicorn companies, where North America is encoded as `1`, through Africa, encoded as `6`.

[147]:
```
Continent_dict={'North America':1,"Asia":2,"Europe":3,"South America":
↪4,"Oceania":5,"Africa":6}

companies['Contnent Number']=companies['Continent'].replace(Continent_dict)

companies
```

[147]:
```
           Company  Valuation Date Joined                         Industry  \
0        Bytedance        180  2017-04-07          Artificial intelligence
1           SpaceX        100  2012-12-01                            Other
2            SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
3           Stripe         95  2014-01-23                          Fintech
4           Klarna         46  2011-12-12                          Fintech
…              …         …          …                              …
1069      Zhaogang          1  2017-06-29  E-commerce & direct-to-consumer
1070   Zhuan Zhuan          1  2017-04-18  E-commerce & direct-to-consumer
```

```
1071     Zihaiguo          1  2021-05-06              Consumer & retail
1072         Zopa          1  2021-10-19                        Fintech
1073        Zwift          1  2020-09-16  E-commerce & direct-to-consumer

               City  Country/Region      Continent  Year Founded Funding  \
0           Beijing           China           Asia          2012     $8B
1         Hawthorne   United States  North America          2002     $7B
2          Shenzhen           China           Asia          2008     $2B
3     San Francisco   United States  North America          2010     $2B
4         Stockholm          Sweden         Europe          2005     $4B
…               …             …             …            …     …
1069       Shanghai           China           Asia          2012   $379M
1070        Beijing           China           Asia          2015   $990M
1071      Chongqing           China           Asia          2018    $80M
1072         London  United Kingdom         Europe          2005   $792M
1073     Long Beach   United States  North America          2014   $620M

                                    Select Investors  Year Joined  \
0     Sequoia Capital China, SIG Asia Investments, S…         2017
1     Founders Fund, Draper Fisher Jurvetson, Rothen…         2012
2     Tiger Global Management, Sequoia Capital China…         2018
3          Khosla Ventures, LowercaseCapital, capitalG         2014
4     Institutional Venture Partners, Sequoia Capita…         2011
…                                                  …          …
1069    K2 Ventures, Matrix Partners China, IDG Capital         2017
1070                           58.com, Tencent Holdings         2017
1071  Xingwang Investment Management, China Capital …         2021
1072  IAG Capital Partners, Augmentum Fintech, North…         2021
1073    Novator Partners, True, Causeway Media Partners         2020

      Years To unicorn   data High Valuation  Contnent Number
0                    5  False           high                2
1                   10  False           high                1
2                   10  False           high                2
3                    4  False           high                1
4                    6  False           high                3
…                  …    …              …               …
1069                 5  False            low                2
1070                 2  False            low                2
1071                 3  False            low                2
1072                16  False            low                3
1073                 6  False            low                1

[1071 rows x 15 columns]
```

Hint

Create a mapping dictionary and use the `replace()` method on the `Category` column. Refer to the example provided above for more information about `replace()`.

1.3.7 Convert `Country/Region` to numeric

Now, suppose that within a given continent, each company's `Country/Region` is given equal importance. For analytical purposes, you want to convert the values in this column to numeric without creating a large number of dummy columns. Use label encoding of this nominal categorical variable to create a new column called `Country/Region Numeric`, wherein each unique `Country/Region` is assigned its own number.

[148]:
```
# Create `Country/Region Numeric` column
# Create numeric categories for Country/Region

companies['Country/Region Numeric']=companies['Country/Region'].
 ↪astype('category').cat.codes
#what we have performed over here is called as the label encoding not one-hot␣
 ↪encoding.

companies.head(2)
### YOUR CODE HERE ###
```

[148]:
```
    Company  Valuation Date Joined                Industry       City  \
0  Bytedance        180  2017-04-07  Artificial intelligence    Beijing
1     SpaceX        100  2012-12-01                    Other  Hawthorne

  Country/Region      Continent  Year Founded Funding  \
0          China           Asia          2012     $8B
1  United States  North America          2002     $7B

                                    Select Investors  Year Joined  \
0  Sequoia Capital China, SIG Asia Investments, S…         2017
1  Founders Fund, Draper Fisher Jurvetson, Rothen…         2012

   Years To unicorn   data High Valuation  Contnent Number  \
0                 5  False           high                2
1                10  False           high                1

   Country/Region Numeric
0                       9
1                      44
```

Hint 1

Review what you have learned about converting a variable with a string/object data type to a category.

Hint 2

To use label encoding, apply `.astype('category').cat.codes` to the `Country/Region` in `companies`.

1.3.8 Convert Industry to numeric

Finally, create dummy variables for the values in the `Industry` column.

[152]:
```
# Convert `Industry` to numeric data
# Create dummy variables with Industry values

### YOUR CODE HERE ###
dummies=pd.get_dummies(companies['Industry'])  #this is also called as the␣
↪one-hot encoding.
dummies
# Combine `companies` DataFrame with new dummy Industry columns
### YOUR CODE HERE ###
```

[152]:
```
      Artificial intelligence  Auto & transportation  Consumer & retail  \
0                           1                      0                  0
1                           0                      0                  0
2                           0                      0                  0
3                           0                      0                  0
4                           0                      0                  0
…                         …                      …                  …
1069                        0                      0                  0
1070                        0                      0                  0
1071                        0                      0                  1
1072                        0                      0                  0
1073                        0                      0                  0

      Cybersecurity  Data management & analytics  \
0                 0                            0
1                 0                            0
2                 0                            0
3                 0                            0
4                 0                            0
…               …                            …
1069              0                            0
1070              0                            0
1071              0                            0
1072              0                            0
1073              0                            0

      E-commerce & direct-to-consumer  Edtech  Fintech  Hardware  Health  \
0                                   0       0        0         0       0
1                                   0       0        0         0       0
2                                   1       0        0         0       0
```

```
3                                      0        0          1           0         0
4                                      0        0          1           0         0
…                                  …      …       …        …       …
1069                                   1        0          0           0         0
1070                                   1        0          0           0         0
1071                                   0        0          0           0         0
1072                                   0        0          1           0         0
1073                                   1        0          0           0         0

      Internet software & services  Mobile & telecommunications  Other  \
0                                0                            0      0
1                                0                            0      1
2                                0                            0      0
3                                0                            0      0
4                                0                            0      0
…                             …                          …    …
1069                             0                            0      0
1070                             0                            0      0
1071                             0                            0      0
1072                             0                            0      0
1073                             0                            0      0

      Supply chain, logistics, & delivery  Travel
0                                       0       0
1                                       0       0
2                                       0       0
3                                       0       0
4                                       0       0
…                                    …     …
1069                                    0       0
1070                                    0       0
1071                                    0       0
1072                                    0       0
1073                                    0       0

[1071 rows x 15 columns]
```

Display the first few rows of `companies`

[157]:
```python
### YOUR CODE HERE ###
companies=pd.concat([companies,dummies],axis=1)
companies
```

[157]:
```
          Company  Valuation Date Joined                          Industry  \
0       Bytedance        180  2017-04-07           Artificial intelligence
1          SpaceX        100  2012-12-01                             Other
2           SHEIN        100  2018-07-03  E-commerce & direct-to-consumer
```

```
3         Stripe        95  2014-01-23                           Fintech
4         Klarna        46  2011-12-12                           Fintech
…           …         …         …                                 …
1069    Zhaogang         1  2017-06-29  E-commerce & direct-to-consumer
1070  Zhuan Zhuan        1  2017-04-18  E-commerce & direct-to-consumer
1071    Zihaiguo         1  2021-05-06              Consumer & retail
1072        Zopa         1  2021-10-19                           Fintech
1073       Zwift         1  2020-09-16  E-commerce & direct-to-consumer

               City  Country/Region      Continent  Year Founded Funding  \
0           Beijing           China           Asia          2012     $8B
1         Hawthorne   United States  North America          2002     $7B
2          Shenzhen           China           Asia          2008     $2B
3     San Francisco   United States  North America          2010     $2B
4         Stockholm          Sweden         Europe          2005     $4B
…             …              …             …           …      …
1069       Shanghai           China           Asia          2012   $379M
1070        Beijing           China           Asia          2015   $990M
1071      Chongqing           China           Asia          2018    $80M
1072         London  United Kingdom         Europe          2005   $792M
1073     Long Beach   United States  North America          2014   $620M

                                        Select Investors  Year Joined  \
0     Sequoia Capital China, SIG Asia Investments, S…         2017
1     Founders Fund, Draper Fisher Jurvetson, Rothen…         2012
2     Tiger Global Management, Sequoia Capital China…         2018
3           Khosla Ventures, LowercaseCapital, capitalG          2014
4     Institutional Venture Partners, Sequoia Capita…         2011
…                                                    …           …
1069    K2 Ventures, Matrix Partners China, IDG Capital          2017
1070                          58.com, Tencent Holdings          2017
1071  Xingwang Investment Management, China Capital …         2021
1072  IAG Capital Partners, Augmentum Fintech, North…         2021
1073    Novator Partners, True, Causeway Media Partners          2020

      Years To unicorn   data High Valuation  Contnent Number  \
0                    5  False           high                2
1                   10  False           high                1
2                   10  False           high                2
3                    4  False           high                1
4                    6  False           high                3
…                   …     …             …              …
1069                 5  False            low                2
1070                 2  False            low                2
1071                 3  False            low                2
1072                16  False            low                3
1073                 6  False            low                1
```

```
      Country/Region Numeric  Artificial intelligence  Auto & transportation  \
0                          9                        1                      0
1                         44                        0                      0
2                          9                        0                      0
3                         44                        0                      0
4                         38                        0                      0
…                        …                       …                     …
1069                       9                        0                      0
1070                       9                        0                      0
1071                       9                        0                      0
1072                      43                        0                      0
1073                      44                        0                      0

      Consumer & retail  Cybersecurity  Data management & analytics  \
0                     0              0                            0
1                     0              0                            0
2                     0              0                            0
3                     0              0                            0
4                     0              0                            0
…                   …            …                          …
1069                  0              0                            0
1070                  0              0                            0
1071                  1              0                            0
1072                  0              0                            0
1073                  0              0                            0

      E-commerce & direct-to-consumer  Edtech  Fintech  Hardware  Health  \
0                                   0       0        0         0       0
1                                   0       0        0         0       0
2                                   1       0        0         0       0
3                                   0       0        1         0       0
4                                   0       0        1         0       0
…                                 …     …      …       …     …
1069                                1       0        0         0       0
1070                                1       0        0         0       0
1071                                0       0        0         0       0
1072                                0       0        1         0       0
1073                                1       0        0         0       0

      Internet software & services  Mobile & telecommunications  Other  \
0                                0                            0      0
1                                0                            0      1
2                                0                            0      0
3                                0                            0      0
4                                0                            0      0
…                              …                          …    …
```

1069	0	0	0
1070	0	0	0
1071	0	0	0
1072	0	0	0
1073	0	0	0

	Supply chain, logistics, & delivery	Travel	Artificial intelligence	\
0	0	0	1	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	
…	…	…	…	
1069	0	0	0	
1070	0	0	0	
1071	0	0	0	
1072	0	0	0	
1073	0	0	0	

	Auto & transportation	Consumer & retail	Cybersecurity	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	
…	…	…	…	
1069	0	0	0	
1070	0	0	0	
1071	0	1	0	
1072	0	0	0	
1073	0	0	0	

	Data management & analytics	E-commerce & direct-to-consumer	Edtech	\
0	0	0	0	
1	0	0	0	
2	0	1	0	
3	0	0	0	
4	0	0	0	
…	…	…	…	
1069	0	1	0	
1070	0	1	0	
1071	0	0	0	
1072	0	0	0	
1073	0	1	0	

	Fintech	Hardware	Health	Internet software & services	\
0	0	0	0	0	

```
1           0         0       0                              0
2           0         0       0                              0
3           1         0       0                              0
4           1         0       0                              0
…       …       …       …                            …
1069        0         0       0                              0
1070        0         0       0                              0
1071        0         0       0                              0
1072        1         0       0                              0
1073        0         0       0                              0

      Mobile & telecommunications  Other  Supply chain, logistics, & delivery  \
0                               0      0                                    0
1                               0      1                                    0
2                               0      0                                    0
3                               0      0                                    0
4                               0      0                                    0
…                           …     …                                 …
1069                            0      0                                    0
1070                            0      0                                    0
1071                            0      0                                    0
1072                            0      0                                    0
1073                            0      0                                    0

      Travel
0          0
1          0
2          0
3          0
4          0
…      …
1069       0
1070       0
1071       0
1072       0
1073       0

[1071 rows x 46 columns]
```

Hint 1

Consider using `pd.get_dummies` on the specified column.

Hint 2

When you call `pd.get_dummies()` on a specified series, it will return a dataframe consisting of each possible category contained in the series represented as its own binary column. You'll then have to combine this new dataframe of binary columns with the existing `companies` dataframe.

Hint 3

You can use `pd.concat([col_a, col_b])` to combine the two dataframes. Remember to specify the correct axis of concatenation and to reassign the result back to the `companies` dataframe.

Question: Which categorical encoding approach did you use for each variable? Why?

[Write your response here. Double-click (or enter) to edit.]

Question: How does label encoding change the data?

[Write your response here. Double-click (or enter) to edit.]

Question: What are the benefits of label encoding?

[Write your response here. Double-click (or enter) to edit.]

Question: What are the disadvantages of label encoding?

[Write your response here. Double-click (or enter) to edit.]

1.4 Conclusion

What are some key takeaways that you learned during this lab?

[Write your response here. Double-click (or enter) to edit.]

Reference

Bhat, M.A. *Unicorn Companies*

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.